

GTRI Interns Go From Classroom Theory to Real-World Impact

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National Intern Day recognizes the contributions interns make to organizations while highlighting the role that meaningful work experiences can play in shaping future careers. While National Intern Day is officially commemorated on July 30, GTRI celebrates its interns and student assistants every day.

Across the Georgia Tech Research Institute (GTRI), interns and student assistants are doing far more than observing from the sidelines. They are developing software, supporting sponsor missions, communicating GTRI's work, exploring emerging technologies, and learning how research moves from an idea to a real-world solution.

GTRI currently has around 350 employees in the "student" classification.

To commemorate the day, students reflected on their experiences and described several common themes: challenging work, supportive mentors, opportunities to take ownership, and a clearer understanding of where their talents might lead.



Justin Allen





Justin Allen

Justin Allen has experienced a combination of technical challenge and practical impact through the [GTRI Research Internship Program \(GRIP\)](#) [3].

A Georgia Tech computer science student working in the Electronic Systems Laboratory's Air National Guard Programs Division, Allen supports the GTTAK project. His work involves integrating live sensor feeds into the Team Awareness Kit, a common operational picture that can improve situational awareness, interoperability, and mission planning for the Georgia National Guard.

Allen helped configure a secure TAK server and integrate more than 4,000 camera feeds. Working with fellow GRIP intern Nikhil Chakre, whose work incorporated nearly 10 additional types of sensor data, the team delivered a functioning system on an accelerated timeline.

Allen said the project gave him something that can be difficult to reproduce in a classroom: a direct connection between technical work and operational purpose.

“Collaborating directly with sponsors to deploy a solution meant for active crisis response gave my work a sense of purpose that a traditional academic environment simply cannot replicate,” he said.

That sense of purpose extended beyond the project itself. Through GRIP tours, Allen visited the Georgia State Patrol's aviation unit, Georgia Emergency Management and Homeland Security Agency headquarters and Dobbins Air Reserve Base. He also encountered technologies ranging from unmanned underwater vehicles and jet engines to an anechoic chamber and a quantum computer.

The breadth of the experience reinforced Allen's interest in computer networking and cybersecurity, while showing him how those skills can contribute to high-impact missions.

His mentor, Mike Curtis, also gave him room to make decisions rather than simply complete assigned tasks.

“He regularly asked for my input on the best path forward and allowed me to take the lead,” Allen said. “This approach empowered me to take ownership of the project and develop autonomy and discipline that I will carry into my future career.”

His advice to future interns is equally direct: Take initiative.

“By taking the initiative to push your limits just a little bit further each time, you will find yourself achieving things you never thought possible a few months ago,” Allen said.



Samiksha Digambar



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Samiksha Digambar, a Georgia Tech industrial engineering student and GRIP intern, has also been trusted with a project that combines emerging technology with a practical operational need.

Working under Christopher Reese, Digambar supports TIDE, an intelligent turnover decision log system designed to streamline maintenance-related queries using artificial intelligence. Her work includes developing a full-stack application with a React front end and FastAPI back end, implementing speech-to-text capabilities and helping create an offline-capable progressive web application for air-gapped maintenance environments.

One of her most significant contributions has been improving the system's search capabilities.

The project began with basic keyword matching, which struggled with abbreviations, specialized jargon and the natural-language questions maintenance technicians might use. Digambar worked with approximately 35,000 rows of data containing inconsistent terminology and domain-specific shorthand.

After testing different methods, she developed a multilayered approach that generates embeddings for semantic similarity searches, uses a customized abbreviation dictionary to normalize queries and employs AI to add relevant context before information is retrieved.

The project changed how Digambar thinks about her own career. Before arriving at GTRI, she was primarily interested in software engineering. Through TIDE, she discovered that she was especially drawn to analyzing data and building tools around the insights it revealed.

"This experience has shifted my career focus toward data engineering and applied AI roles," she said.

Just as importantly, she learned how to navigate the realities of a large research organization.

"In a classroom, where everything is handed to you upfront, it becomes difficult to learn how to navigate processes and troubleshoot effectively," Digambar said. "GTRI teaches you how to do that."

She encourages future interns to arrive ready to learn, ask questions, and take responsibility for their work.

"GTRI gives you real responsibility and trusts you to deliver, which is both challenging and incredibly rewarding," she said.



Graham Shepard



Graham Shepard

Graham Shepard, a Georgia Tech mechanical engineering student, serves as a student assistant in the [SEEDLab](#) makerspace, where he helps research engineers prototype and fabricate components for projects using a variety of specialized manufacturing equipment.

For Shepard, the most rewarding part of the position is helping others succeed.

“The most meaningful part of my job at GTRI is getting to help others with their projects on a daily basis and leaving a lasting impact by teaching them how to operate different tools in the lab,” he said.

Among the projects he is most proud of was helping design and fabricate commemorative plaques for the STEM@GTRI intern class using the SEEDLab laser cutter. The plaques recognized the interns’ accomplishments while celebrating GTRI’s commitment to STEM outreach.

“It made me proud to contribute to the reinforcement of STEM outreach and interest,” Shepard said. “The supervisors shouted me out at the ceremony for my help.”

Working in SEEDLab has also allowed Shepard to develop practical manufacturing skills that extend well beyond the classroom. He has gained experience operating CNC milling equipment, resin printers, 3D printers, and other fabrication tools while supporting real research efforts.

His supervisor has also encouraged his professional growth by introducing him to additional opportunities across GTRI, helping him better understand the breadth of work taking place throughout the organization.

The experience has reinforced Shepard’s interest in mechanical design and advanced manufacturing.

“This opportunity reinforced my interest in designing and fabricating parts to solve problems,” he said.

Asked what advice he would offer future student employees, Shepard answered with a smile: “They have donuts in the kitchen in the Baker Building on Fridays.”

More seriously, Shepard summed up his internship experience by simply saying: “Making a difference.”



Not every GTRI internship centers on developing a technical system. Ava James and Saige Forbes have spent the summer helping GTRI communicate its work and strengthen its connections with internal and external audiences.

Saige Forbes



Saige Forbes

Saige Forbes, a Harvard University student studying economics and computer science, works as a student assistant with GTRI's Communications unit. One of her most meaningful assignments grew directly from her own experience applying to GTRI as a student from outside Georgia Tech.

She found relatively few resources explaining what a GTRI student position might involve. This summer, she has helped address that gap by contributing to a student employment and retention campaign designed to make opportunities clearer and more accessible to future applicants.

Forbes has also supported GTRI's cybersecurity communications campaign, exploring alternatives to email for sharing security information. Potential approaches include WebWISE content, social media posts, and digital displays in GTRI facilities.

The work has helped her see how individual assignments connect to a larger institutional purpose.

"Although each person has their own responsibilities and tasks, everyone is working toward the same goal, which is to be the voice of Georgia Tech Research Institute to the organization and to the world," Forbes said.

She describes the experience, relationships, and opportunities that strengthened her skills as unforgettable.

"To a student who is considering embarking on their journey with GTRI, I would reassure them that it will only do them good," she said.



Ava James



Ava James

Ava James, a University of Georgia entertainment and media studies student, supports external communications as a communications student assistant. Her work involves helping ensure that GTRI reaches the appropriate

audiences with clear and accurate messages, particularly as the organization builds and maintains relationships with government sponsors.

For James, one of the most valuable lessons has been seeing how a professional communications team functions in practice.

“It is one thing to learn about corporate workflow in a classroom, but it is something entirely different to experience it yourself,” she said.

She described herself as a hands-on learner and said observing the daily coordination, review and collaboration behind GTRI communications has given her a stronger understanding of professional teamwork.

James also credited her mentors with providing thoughtful feedback that she can apply beyond a single assignment.

“They are amazing at giving feedback on my projects, and I am thankful to be able to take that feedback and implement it into my future work,” she said.

Her advice to students considering a position at GTRI is to learn about the organization before arriving.

Researching GTRI's mission, accomplishments, and contributions helped her understand the environment and begin her internship with greater context and confidence.



GTRI's connections to students extend further down, into high school as well. Georgia high school students get invaluable exposure to STEM-related work, which helps them solidify their future collegiate pursuits as well as their career aspirations. The hands-on experience they get at GTRI puts them ahead of their peers.

Falyssa Jade Ly



Falyssa Ly

For Falyssa Ly, a second-time participant in the [STEM@GTRI](#) ^[10] high school internship program, that path has led her into quantum computing.

Ly worked in the CIPHER Laboratory's Quantum Systems Division under mentor Bryan Gard, Ph.D. During her first summer, she studied quantum error correction, including how rotated surface codes can help identify and manage errors in quantum systems. She simulated the use of a rotated surface code on an entangled Bell state and examined how increasing code distance affected fidelity under simulated noise.

Her work required both technical understanding and the ability to make highly complex ideas understandable. To illustrate how errors moved through the system, Ly transformed one of her visualizations into an animated GIF.

The result became one of the proudest moments of her internship.

“My 63-year-old grandma was in the audience that time, and afterward she came up to me and told me she understood what I was presenting,” Ly recalled.

For Ly, the moment carried meaning beyond the technical accomplishment. Her grandmother, who speaks limited English, left school in Mexico to help support her family and later immigrated to the United States as a teenager. Seeing her granddaughter explain quantum computing at age 17 represented what Ly described as a full-circle moment for her family.

“It served as a testament to my abilities as a researcher by showing that what I did could teach someone science despite a language and educational barrier,” she said.

This summer, Ly's group turned its attention to quantum-computing architectures, including superconducting systems, neutral atoms and trapped ions. Her contribution involved creating an interactive website explaining how quantum computers operate alongside classical computers and supercomputers.

The experience has helped Ly, an incoming University of Georgia computer engineering student, envision a future that combines computer engineering, physics and assistive robotics. It has also shown her the value of a workplace where a young researcher can ask questions, contribute ideas and feel taken seriously. “Working at GTRI has meant being shown what democratized science truly is,” Ly said, “and being surrounded by some of the brightest, kindest and most passionate minds choosing to do good for the benefit of society.”



Emma Catherine Lane



Emma Lane

Emma Lane, an AP Honor student at Chamblee High School, spent her summer exploring the complete engineering design process as a STEM@GTRI high school intern under the mentorship of Erick Maxwell, Ph.D. Working in electrical engineering, Lane followed a project from initial concept through final fabrication, learning to design electrical circuits, develop printed circuit board (PCB) layouts, and solder electronic components. Those skills came together in a final project aimed at addressing a real-world medical challenge: a humidity sensor designed to detect moisture buildup inside orthopedic casts before prolonged exposure can lead to skin irritation or infection.

Lane designed and built the device before presenting her work to GTRI researchers and mentors at the conclusion of the program.

The experience gave her a firsthand look at how engineering can improve people's lives while introducing her to the collaborative nature of research.

Throughout the summer, Lane also learned from a variety of GTRI professionals, including Frank Nordel and Rebekah Price, who taught soldering fundamentals; Jon Duke, Ph.D., who introduced students to medical informatics; and Samantha Lie Tjauw, who helped the interns strengthen their public speaking skills.

She also credited STEM@GTRI educators Lucretia Gant and Michael Matthews, along with the college interns who worked alongside the high school students, for creating an environment that was both supportive and challenging.

One of the summer's unexpected highlights was sharing the experience with her cousin, who was selected for the same intern cohort.

"Having the opportunity to learn and grow alongside him made this experience even more special," Lane wrote.

As she looks toward college, Lane said the internship reinforced her interest in biomedical and electrical engineering and provided practical experience that will help shape her future studies.

"I'm excited to apply these skills as I continue pursuing opportunities in biomedical and electrical engineering," she said.



The students' experiences show that an effective internship is not defined simply by proximity to interesting work. It depends on whether students are invited to participate in that work, trusted with meaningful responsibilities and supported as they learn.

At GTRI, that can mean translating quantum error correction for a public audience, building mission-ready situational awareness tools, improving an AI-enabled maintenance system or helping the organization communicate more clearly with the people it serves.

For the interns, the summer provides experience, technical skills and professional connections. For GTRI, it brings new perspectives, energy and talent into the research enterprise. And for both, it offers a glimpse of what can happen when students are given the opportunity not merely to watch the future being built, but to help build it.

Writer: Christopher Weems

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